SIP and carriers

Name the path. G.711 is the default you can cite. Studio ports are not every trunk.

1. **Three ways a call can enter** The technical requirements describe three SIP stories. Five9 can provide ITSP trunks. Your ITSP can connect to Five9 session border controllers, and there is a checklist for that. Or a gateway you already own — Avaya, Cisco, or Asterisk — can talk to Five9 over the internet, a VPN, or Direct Connect.
2. **DNS, codecs, and the default voice** The requirements prefer DNS SRV for the SIP targets. G.711 is the default codec, and the PDF describes it as uncompressed. Do not invent a packetization time. A 20 millisecond ptime shows up in practitioner notes about Cisco gateways. It is not a number this lab will attribute to the Five9 PDF.
3. **How much bandwidth a seat costs** The requirements give planning numbers. Agent Desktop Plus and the Plus adapters are 128 kbps. The Java agent is 88 kbps. Agent Assist adds about 12 kbps. A VPN adds about 64. CRM or other business apps add about 64. A UC client such as Teams, Zoom, Office@Hand, RingCentral, or Nextiva is called out around 320 kbps.
4. **NAT, idle timers, and split tunnels** NAT and PAT are acceptable. Double NAT is not. UDP bindings that idle out cause one-way audio. The requirements use a 30 second UDP idle timeout as the planning number, and they tell you to send keepalives when the binding is shorter.
5. **Direct Connect and the edges** Direct Connect in the requirements is a Layer-3 handoff. There is no Layer-2 requirement. The port is 10, 100, or 1000 full duplex, and failover uses BGP. Equinix Fabric edges named in the PDF this lab read are Chicago CH3, Washington DC10, London LD5, Amsterdam AM4, Frankfurt FR8, São Paulo SP4, Tokyo TY11, and Sydney SY5.
6. **TLS, and ports that are not universal** Unencrypted APIs are not the story. TLS 1.2 is required. TLS 1.1 is not supported. The requirements also list TLS 1.3 inbound ciphers: AES-128-GCM, AES-256-GCM, and CHACHA20.
7. **Headsets, and the call that only fails one way** USB noise-canceling headsets are the recommendation. Analog headsets are not supported. Wireless is not recommended once you are above roughly ten agents. Windows 11 Efficiency Mode can drop live calls. Java admin is not the Mac ARM path; use the web admin there. None of that is a SIP trunk, and all of it still stops a call.

Confirm before relying.



A note from the student behind this lab

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